

Assignment 5: LSB steganography

Task 1. LSB extraction and re-insertion

- 1.1. Read a color image and convert it to grayscale
`I = imread('filename'), G = rgb2gray(I);`
- 1.2. Extract the k=8 bit-planes of the image
For each pixel I(i,j) of the image, convert it to binary (dec2bin, de2bin)
Store each bit of the binary value in the corresponding matrix
- 1.3. Display each bit-plane of the image
`figure, imshow(bitplane(:, :, k), []);`
`figure, subplot(a,b,c), imshow(bitplane(:, :, k))`
- 1.4. Reconstruct the grayscale image
For each pixel of each bit-plane bitplane(i,j,k), convert the binary value to decimal and store it into a new matrix (bin2dec, bi2dec)
- 1.5. Save the modified image
`imwrite(G_mod, 'format', parameters);`

Task 2. Embedding a message into the LSB

- 2.1. Read a color image and convert it to grayscale
`I = imread('filename'), G = rgb2gray(I);`
- 2.2. Extract the kbit-planes of the image
`bitget(G,k);`
- 2.3. Convert a string into a binary message of length L
`repmat(s,M,N), dec2bin()`
- 2.4. Insert the message into random positions of the LSB
`rand('state',key), randperm()`
`payload = log2(length(s))-length(G(:));`
- 2.5. Reconstruct the image and save it
- 2.6. Extract the LSB of the modified (stego)image
Compute the L positions as in embedding by using same key
Compare with the input message: `xor(message_or, message_rec)`
- 2.7. Compare the cover image x and the modified stego image y
Pixel-wise difference:
`diff = abs(x-y);`
Distortion:
`d(x,y) = sum(sum((x-y).^2));`
Peak Signal-to-Noise Ratio (PSNR):
`[M,N] = size(x);`
`mse = sum(sum((x-y).^2))/(M*N);`
`p = 20 * log10(255) - 10 * log10(mse);`
Structural SIMilarity index (SSIM):
`[ssim, map] = SSIM(double(x), double(y));`

Task 3. Embedding of binary image into LSB

- 3.1. Read a color cover image of size [M,N]

- 3.2. Read a grayscale secret image of size [M,N] and make it binary
`S_bw = im2bw(S); or S_bw = S>128;`
- 3.3. Extract the channel (e.g. green) where the message is embedded
- 3.4. Get the LSB plane of the green channel
`LSB_green = bitget(green,1);`
- 3.5. Replace the LSB with the secret image
`LSB_green = S_bw;`
- 3.6. Reconstruct the green channel `green_mod`
- 3.7. Reconstruct the stego color image and save it in TIFF format
`stego = cat(3, I(:, :, 1), green_mod, I(:, :, 3) );`
`imwrite(I_stego, 'stego.tiff', 'tiff', 'Compression', 'none');`
- 3.8. Compare each channel of the cover image `x` and the modified stego image `y`
Pixel-wise difference
Distortion
Peak Signal-to-Noise Ratio (PSNR)
Structural SIMilarity index (SSIM)
Color difference: $\text{delta} = \sqrt{(\text{Lx}-\text{Ly})^2 + (\text{Ax}-\text{Ay})^2 + (\text{Bx}-\text{By})^2}$;
- 3.9. Repeat the embedding procedure on a different channel (red, blue) and compute again the above metrics. Indicate the difference
- 3.10. Modify the algorithm in such a way that the secret image is scrambled according to a key (see Task 2)
- 3.11. Modify the algorithm so that a smaller image can be embedded
- 3.12. Recover the hidden image
Extract the LSB of the modified (stego) image
Compare with the input image
`xor(message_or, message_rec)`

Task 4. Embedding of binary image into LSB (robustness)

- 4.1. Explore the effects of various processing on LSB stego images
- 4.2. Verify that the secret message (image) is inside the stego image
- 4.3. Crop an arbitrary portion of the image and verify what happens to the LSB message.
What if the message was scrambled at embedding?
`I_crop = imcrop(I,[x,y,h,w]);`
- 4.4. Add some noise to the stego image and verify what happens to the LSB message
`I_noise = imnoise(I_stego, 'noise_type', params);`
- 4.5. Compress the image with JPEG and verify what happens to the LSB message
`imwrite(I_stego, 'stego.jpg', 'jpeg', 'Quality', 100);`
`I_jpeg = imread('stego.jpg');`

Task 5. Embedding of binary image into LSB (detection)

- 5.1. LSB embedding leaves traces into image histogram
- 5.2. Read the color stego image from Task 3, extract the green channel hosting the LSB secret message
- 5.3. Compute the pixel histogram of the original green channel and of the stego green channel
`h_or = hist(green_or, 256);`

```
h_stego = hist(green_stego, 256);
```

5.4. Plot the histograms and observe their differences

```
figure, bar( h_or );
```

```
figure, bar( h_stego );
```

5.5. LSB revealed by LSB enhancement

```
figure, imshow(255*bitget(I_stego,1));
```